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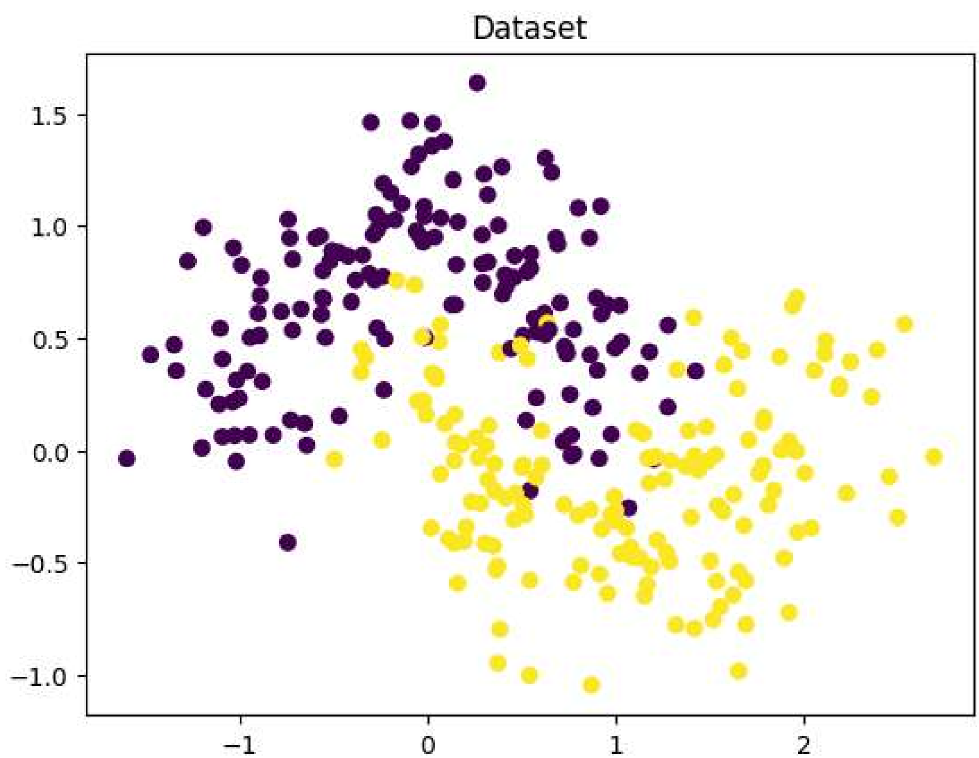
###ROLL NO: 18 BATCH: A1

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_moons
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix
```

In [2]:
```
#Create 2D dataset
x, y = make_moons(n_samples=300, noise=0.25, random_state=42)
plt.scatter(x[: , 0], x[: , 1], c = y)
plt.title("Dataset")
plt.show()
```



In [5]:
```
#Train-test split
x_train, x_test, y_train, y_test = train_test_split(
    x,y,test_size = 0.3,random_state = 0)
```

In [4]:
```
#Scaling (needed for LR amd KNN)
scalar = StandardScaler()
x_train_scaled = scalar.fit_transform(x_train)
x_test_scaled = scalar.transform(x_test)
```

In [14]:

```
#Part A Logistic Regression
from sklearn.linear_model import LogisticRegression
lr = LogisticRegression()
lr.fit(x_train_scaled, y_train)
y_pred_lr = lr.predict(x_test_scaled)
print("Logistic Regression")
print("Accuracy: ", accuracy_score(y_test, y_pred_lr))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_lr))
```

```
Logistic Regression
Accuracy:  0.8777777777777778
Confusion Matrix:
 [[34  5]
 [ 6 45]]
```

In [11]:

```
#Part B Decision Tree
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(max_depth = 4)
dt.fit(x_train, y_train)
y_pred_dt = dt.predict(x_test)
print("Decision Tree")
print("Accuracy: ", accuracy_score(y_test, y_pred_dt))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_dt))
```

```
Decision Tree
Accuracy:  0.8555555555555555
Confusion Matrix:
 [[30  9]
 [ 4 47]]
```

In [12]:

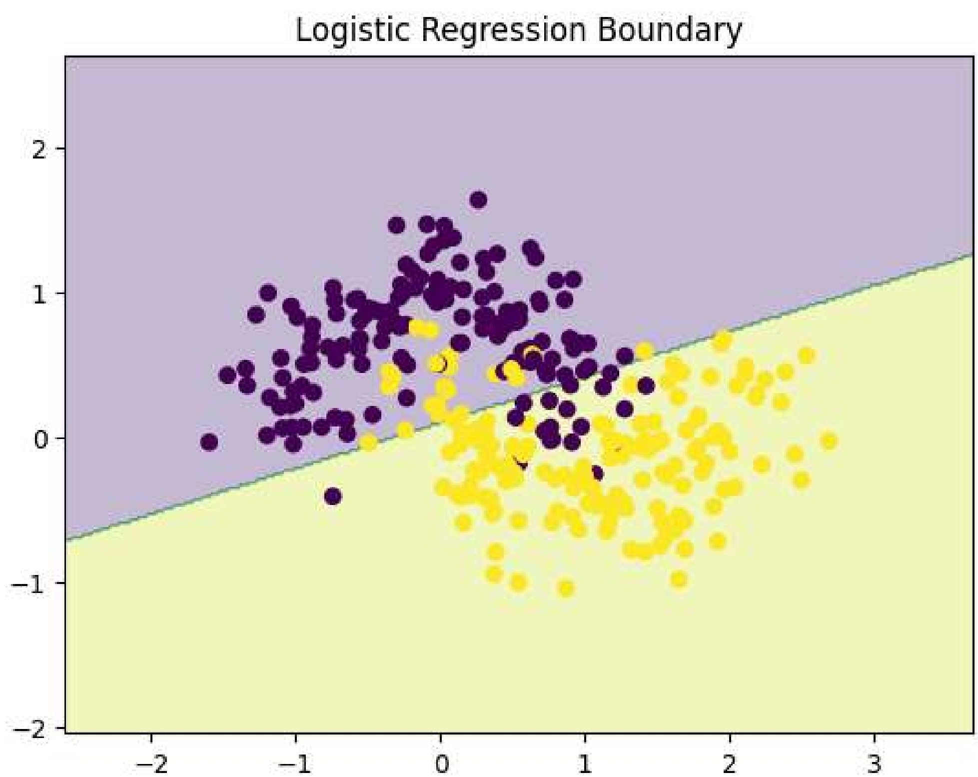
```
#Part C KNN
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors = 5)
knn.fit(x_train, y_train)
y_pred_knn = knn.predict(x_test)
print("KNN")
print("Accuracy: ", accuracy_score(y_test, y_pred_knn))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_knn))
```

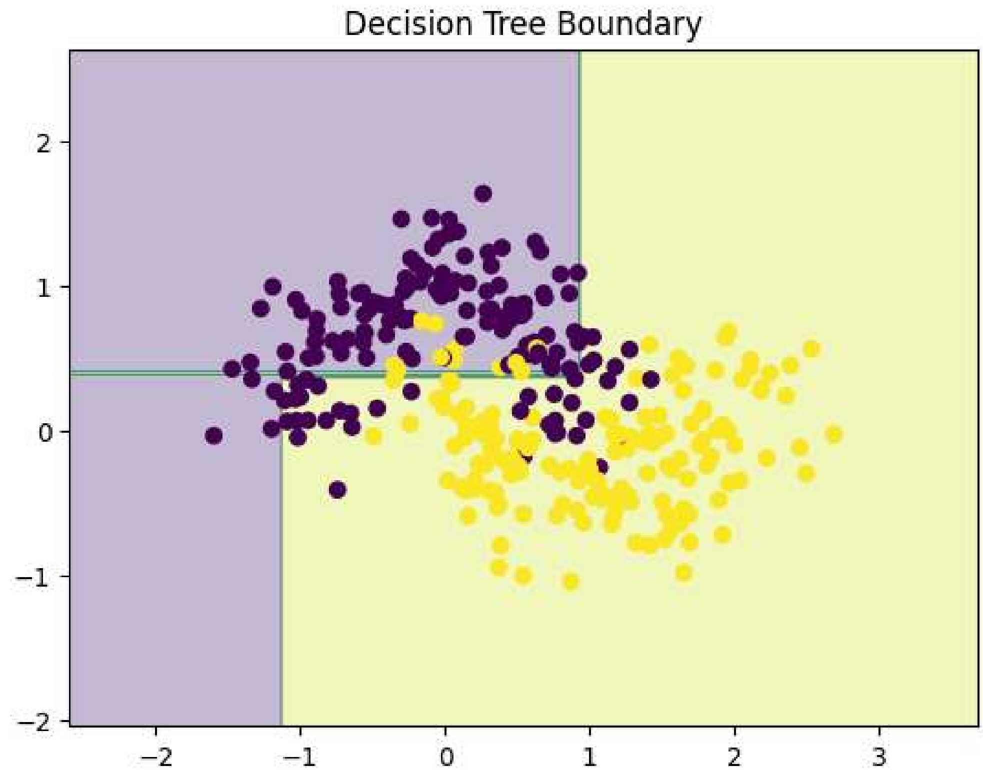
```
KNN
Accuracy:  0.9111111111111111
Confusion Matrix:
 [[38  1]
 [ 7 44]]
```

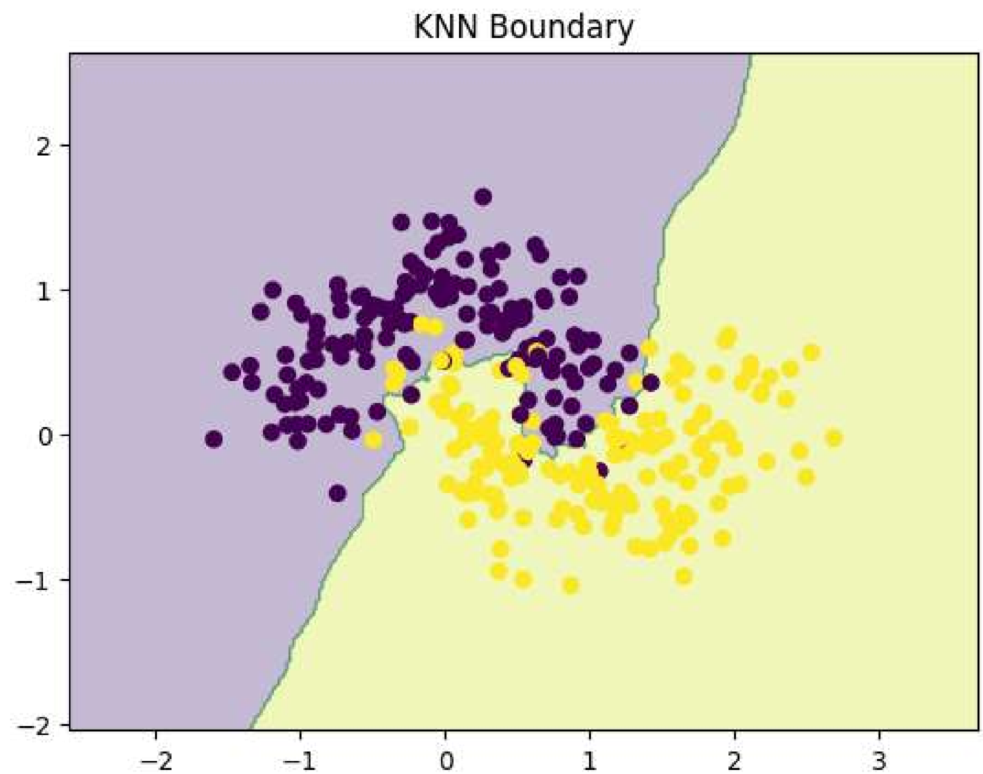
In [25]:

```
#Part E Visualization
def plot_boundary(model, scaled, title):
  h = 0.02
  x_min, x_max = x[:, 0].min() - 1, x[:, 0].max() + 1
  y_min, y_max = x[:, 1].min() - 1, x[:, 1].max() + 1
  xx, yy = np.meshgrid(np.arange(x_min, x_max, h), np.arange(y_min, y_max,
h))
  grid = np.c_[xx.ravel(), yy.ravel()]
  if scaled:
    grid = scalar.transform(grid)
  z = model.predict(grid)
  z = z.reshape(xx.shape)
  plt.contourf(xx, yy, z, alpha = 0.3)
  plt.scatter(x[:, 0], x[:, 1], c = y)
  plt.title(title)
  plt.show()

plot_boundary(lr, True, "Logistic Regression Boundary")
plot_boundary(dt, False, "Decision Tree Boundary")
plot_boundary(knn, False, "KNN Boundary")
```




Decision Tree Boundary
2
1
0
−1
−2
−2
−1
0
1
2
3


KNN Boundary
2
1
0
−1
−2
−2
−1
0
1
2
3